Outliers and Statistical Approaches in Data Mining

-Finding the "Odd One Out"

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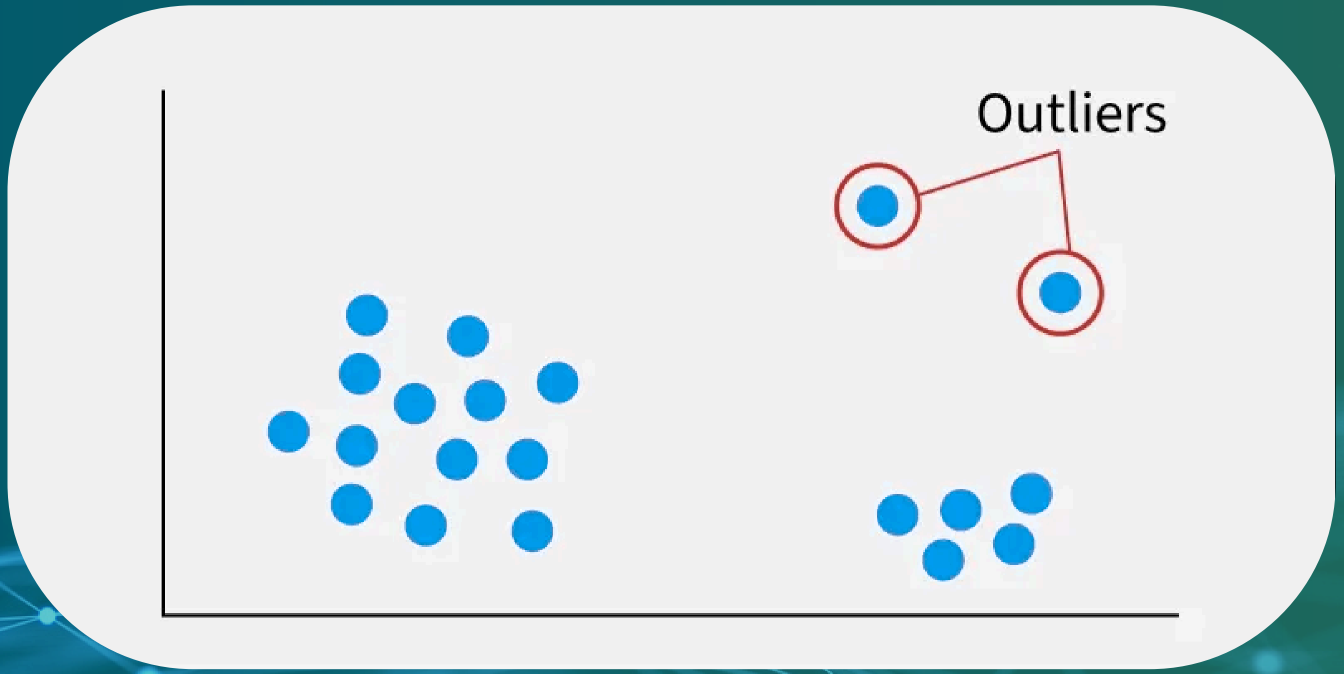
INTRODUCTION TO OUTLIERS



- An outlier (or anomaly) is a data observation that deviates significantly from the rest of the data.

Why Do They Matter?

- They can be errors that need cleaning (e.g., a data entry mistake, sensor malfunction).
- They can be critical events that need attention (e.g., a fraudulent transaction, a network intrusion, a medical symptom).

The Main Challenge:

- How do we distinguish between meaningless noise and an important, actionable outlier?

OUTLIER DETECTION METHODS

Unsupervised Methods

- The most common approach. No labels are required.
- Assumes that outliers are rare and isolated compared to normal data points.
- Example: In a cluster analysis, points that do not belong to any cluster or are far from their cluster's center can be flagged as outliers.

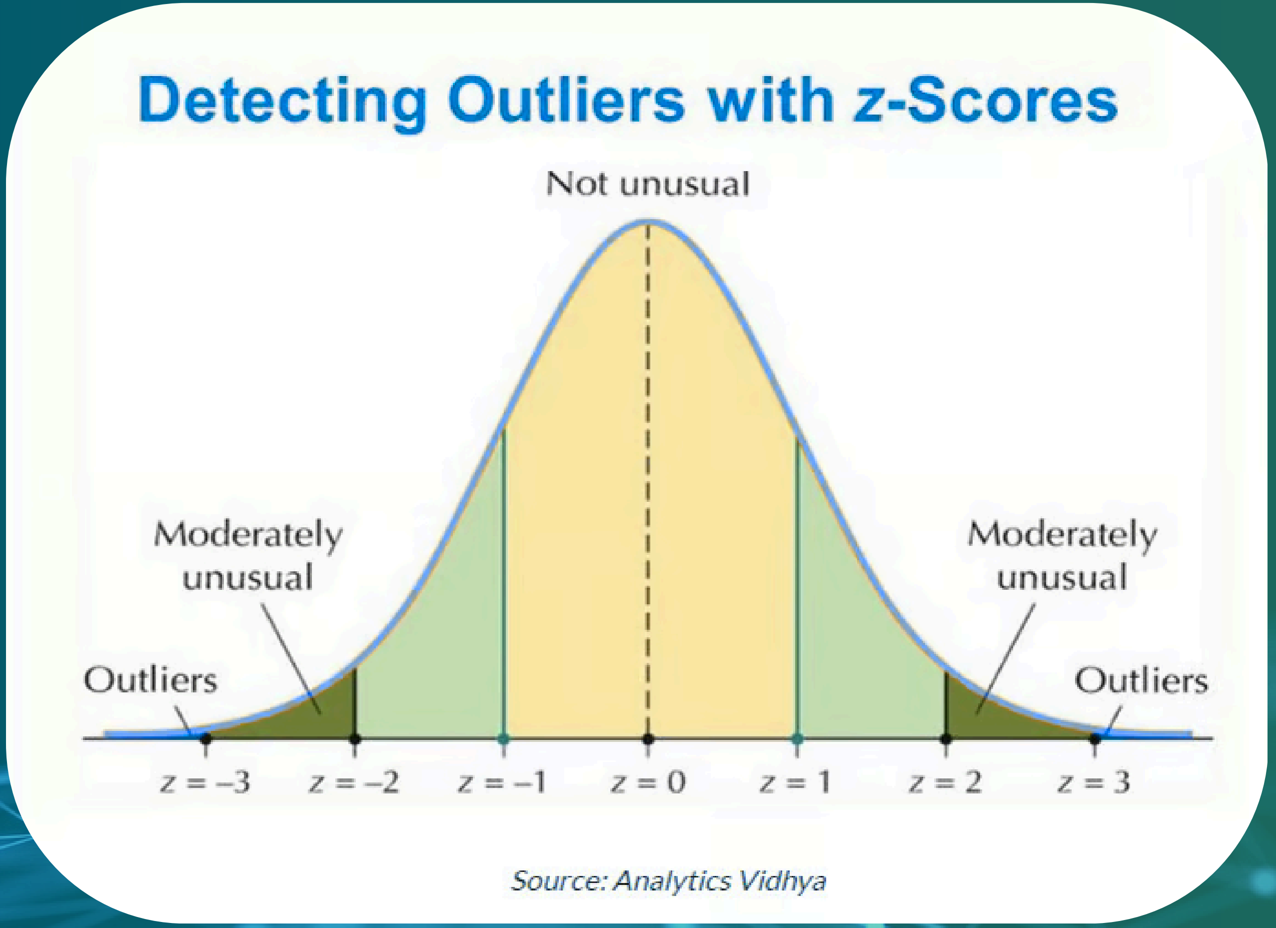
Supervised Methods

- Requires a training dataset with pre-labeled "normal" and "outlier" data.
- Trains a classifier to distinguish between them.
- Challenge: Labeled outlier data is often very rare and expensive to obtain.

Semi-Supervised Methods

- A practical middle ground. The model is trained only on normal data.
- It learns to recognize what is "normal."
- Any data that does not fit this model is then flagged as an outlier.

STATISTICAL APPROACHES TO OUTLIER DETECTION



Using Probability to Identify Anomalies

- Assume the "normal" data points follow a known statistical distribution (e.g., a bell curve).
- How It Works:
  - Fit a statistical model to your dataset.
  - Calculate the probability of each data point occurring under this model.
  - Points with a very low probability are declared outliers.

- Common Technique: Z-Score
  - For data that follows a Normal (Gaussian) distribution.
  - The Z-score measures how many standard deviations a data point is from the mean.
  - A common rule of thumb is to flag anything with a Z-score above 3 or below -3 as an outlier.

APPLICATION: INTRUSION DETECTION & FINANCIAL FRAUD

OUTLIER DETECTION IN SECURITY

INTRUSION DETECTION

- First, model the characteristics of normal network traffic (e.g., typical data volume, protocols used, user login times).
- An outlier detection system then monitors the network in real-time.
- Events that deviate significantly from the normal pattern (e.g., a massive data transfer at 3 AM) are flagged as potential intrusions.

FINANCIAL FRAUD

- The system learns your personal spending patterns (e.g., typical purchase amount, locations, types of stores).
- A transaction that is an outlier to your personal profile (e.g., a large purchase made in another country) is immediately flagged for review.

APPLICATION: RECOMMENDER SYSTEMS & FINANCIAL ANALYSIS

FINDING VALUE IN ANOMALIES

RECOMMENDER SYSTEMS

- Detecting unusual rating patterns can identify fraudulent reviews or "shilling attacks" meant to manipulate the system.
- Identifying users with very unique, outlier tastes can help provide them with specialized, non-mainstream recommendations.

FINANCIAL ANALYSIS

- Anomalies in stock trading data can signal market manipulation or a significant impending event.
- Outliers in a company's financial statements (e.g., unusually high revenue from a small division) can flag areas that require a deeper audit.

THANK YOU